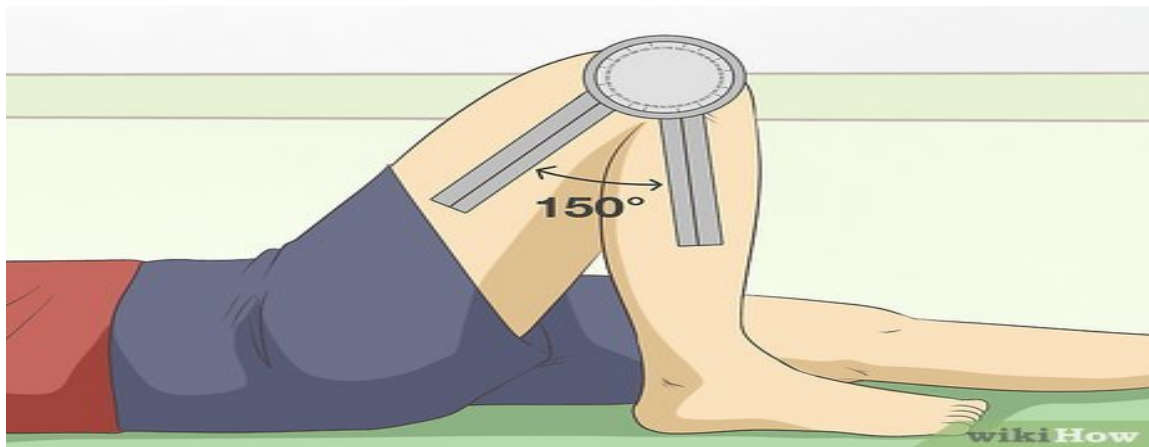


Heel slides target **knee flexion**, the bending motion your knee needs for daily activities.

Each rep gently pushes the joint further into a bend, gradually restoring its functional range. Over time, this translates to better movement when walking, sitting, and climbing stairs.

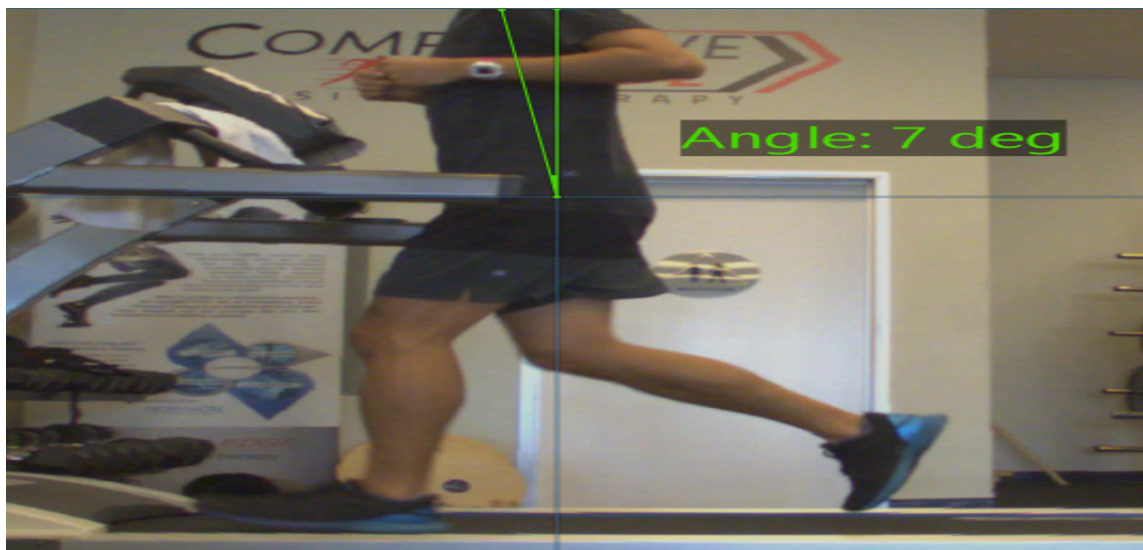
Heel slides apply a gentle, repeating load that encourages the joint and soft tissue to stretch a little more with each rep. (Knee flexion is our tracking angle)

- 1) **Knee flexion** is the movement of bending your knee to decrease the angle between your thigh and lower leg. A healthy adult knee typically has a functional active range of motion between 0 degree and 120 degree and 140 degree of flexion.



- 2) **Trunk Lean Angle** (Torso leaning forward)

Trunk lean refers to how far your torso (trunk) is pitching forward as you run. More specifically, forward lean is measured relative to your torso's usual vertical position (such as when you're standing up straight).



Leaning forward at the hip, even if only a little bit, can improve your hip's overall range of motion (ROM). Not only does this allow for better mobility at the joint, but having sufficient hip ROM is also an imperative facet for achieving full hip extension.

The human body functions as an interconnected kinetic chain. When one joint, muscle group, or body segment is unable to perform its intended function effectively, other structures compensate to maintain movement. Over time, these compensatory mechanisms can place excessive stress on surrounding joints, muscles, and tissues, potentially leading to pain, reduced performance, or further injury.

Many runners with limited ROM at the hip wind up arching their lower back in order to achieve that necessary extension. As a result, the lumbar spine undergoes increased pressure, and the remainder of the kinetic chain is less capable of properly absorbing shock. All of this combined ultimately leads to issues with low back pain and increased risk of stress fracture.

A similar effect occurs when your glutes are inactive. Because your gluteal muscles are such a prominent source of power generation, when they aren't functioning at peak capacity, your other body parts have to take on extra work that they weren't built to handle.

Which brings us to one of the biggest (and most surprising) impacts of insufficient trunk lean: [its correlation to excessive loading at the knee joint](#).

If your glutes aren't active, your body will resort to the next "best" alternative: the quadriceps. Because the quads are directly connected to your knee cap, when the muscles are overworked and taking on a majority of the loading forces, those same forces are transferred right to the knee joint. After prolonged periods of running with increased stress at the joint, you're more susceptible to longer-term running injuries, like runner's knee and patellar tendonitis.



3) Ankle Dorsiflexion Angle

This is open chain dorsiflexion - that is when the ankle is not touching the ground and you try to move your ankle up towards your shin. This angle is usually around 17 degrees.



Ankle Range of Motion

Ankle Range of Motion testing is divided into active range of motion (AROM) and passive range of motion (PROM), with overpressure being superimposed at the end of available range to assess the end feel.

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Then there is something called closed chain dorsiflexion - that is when the ankle/ foot is firmly pressed against the floor and you try to bend your knee without lifting your feet up.

